

Task Sheet — Data Analysis and Validation

Urban Cycle Hire Scheme | Full Year Performance Review 2024

Business Question

Which stations and time periods are underperforming in hire volume and revenue across the GreenWheels network in 2024, and what does the full-year data reveal about overall scheme performance, seasonal patterns, and areas requiring operational attention?

File Provided

File	Contents
urban_cycle_hire_aggregated.xlsx	Aggregated quarterly hire data for 10 GreenWheels stations across 2024 (40 records). Includes Total Hires, Average Hire Duration, Total Revenue, Active Bikes, and Customer Satisfaction Score. The dataset has been pre-merged and is ready to analyse.

Open the dataset and cross-check its integrity before any analysis begins.

Task 1 — VALIDATE

- 1. Check row count and column structure**
Confirm there are 40 data rows and all 10 columns are present and correctly named.
- 2. Check for blanks**
Use ISBLANK or conditional formatting to identify any empty cells across all columns.
- 3. Apply conditional formatting to surface anomalies**
Highlight numeric columns to flag unusually high or low values; apply a colour scale or icon set.
- 4. Sort each numeric column and inspect MIN and MAX**
Use MIN, MAX and sorting to find outliers in Total Hires, Revenue and Active Bikes.
- 5. Document every issue found**
On a separate notes sheet or printed sheet, record: what you found, where it is, and what action you took.

Use Excel formulas to answer the business question with evidence from the data.

Task 2 — SUMMARISE

- **1. Calculate overall totals**
Use SUM and COUNT to find total hires, total revenue and number of station-quarters across the dataset.
- **2. Calculate averages, minimum and maximum**
Use AVERAGE, MIN and MAX for Total Hires and Total Revenue across all records.
- **3. Use COUNTIF and SUMIF to analyse by station and zone**
E.g. total hires for Central zone; number of quarters where hires exceed 400.
- **4. Use AVERAGEIF to compare average revenue by zone**
Identify which zones generate the highest and lowest average revenue per quarter.
- **5. Answer the business question**
Write a formula-backed statement identifying the underperforming stations and periods.

Build a Pivot Table to summarise the data across multiple dimensions.

Task 3 — PIVOT

- **1. Insert a Pivot Table on a new sheet**
Select all data including headers before inserting. Name the sheet 'Pivot Analysis'.
- **2. Summarise Total Hires and Total Revenue by Station Name and Period**
Place Station Name in Rows, Period in Columns, and both numeric fields as Values.
- **3. Add a second Pivot Table summarising by Zone**
Show total hires and average satisfaction score per zone to identify zone-level patterns.
- **4. Insert a Pivot Chart**
Choose a chart type appropriate to the question: line chart for quarterly trend, bar chart for station comparison.
- **5. Format the chart clearly**
Add a descriptive title, label the axes and remove unnecessary gridlines or legend entries.

Build a clean, self-contained dashboard on a new sheet.

Task 4 — PRESENT

- **1. Create a new sheet called 'Dashboard'**
Clear any default content and set a clean background.
- **2. Place your Pivot Chart on the Dashboard sheet**
Copy and paste as a picture or move the chart to this sheet.
- **3. Add at least two key summary figures**
Display total annual hires and total annual revenue as prominent headline figures with clear labels.
- **4. Add at least one Slicer**
Connect a Slicer for Zone or Period so the dashboard can be filtered interactively.
- **5. Check layout and readability**
Ensure all text is legible, figures are clearly labelled and the sheet would make sense to a non-technical reader.

Write a structured summary of what your analysis shows.

Task 5 — INTERPRET

■ 1. Identify at least two trends

Reference specific numbers: e.g. which station grew most across the year, which zone consistently outperformed.

■ 2. Identify at least one anomaly

Describe the anomaly, where it appears in the data, and what you think may have caused it.

■ 3. State what the data does support

Make a clear evidence-based statement about scheme performance backed by your formula outputs.

■ 4. State what the data does not support

Identify at least one conclusion that cannot be drawn from this dataset alone and explain why.

■ 5. Recommend one action

Based on your findings, state one operational recommendation for the GreenWheels management team.

Extension Tasks (Optional — for stronger learners)

These tasks are not required to pass. They demonstrate deeper analytical thinking and technical skill.

■ E1. Calculate the median for Total Hires and Total Revenue

Use the MEDIAN function and compare the result to the AVERAGE. What does a significant difference tell you about the distribution of values? Write two sentences explaining what it suggests.

■ E2. Add a trendline to your Pivot Chart

Right-click your data series, select Add Trendline, and choose Linear. Describe in one sentence what the trendline suggests about future performance if the current pattern continues.

■ E3. Record a macro for conditional formatting

Use Developer > Record Macro to capture the steps of applying your conditional formatting rules. Give the macro a clear name and test that it runs correctly on a copy of the dataset.

Reflection

Answer the following questions in writing after completing all five tasks.

1. What validation steps did you carry out, and why are they important before analysis begins?
2. What anomalies did you find in the dataset, and what decisions did you make about how to handle them?
3. How confident are you in your findings? What additional data or checks would increase that confidence?

Criteria evidenced in this demonstration: K7 (Automation) | K8 (Filtering) | K9 (Statistical methods) | K10 (Data quality issues) | K11 (Validation methods) | S7 (Trends and patterns) | S8 (Cross-checking) | S9 (Audit results) | S16 (Clean data) | B2 (Independence) | B3 (Initiative) | B4 (Organised approach)